

Memory

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Chapter Overview

This chapter provides a discussion of memory in philosophy and neuroscience, with each section organized around three questions: (1) What is memory? (2) What is remembering? (3) What are memories? Each section begins by exploring ways that memory has been taxonomized, both by scientists and by philosophers of memory, emphasizing the kinds that have been particularly relevant for discussions at the intersection of these disciplines. The focus moves next to the mental act of remembering—the processes that are involved in our capacity to encode, store, and retrieve information—and the mechanisms that make remembering unique but also similar to other cognitive operations. Finally, each section draws attention to the individual mental states we call “memories,” regardless of whether they involve memory traces or representations, and what the nature of such entities may be. Throughout, we make note of the ways in which philosophical and neuroscientific work on memory have been pursued independently and discuss the current trend toward interdisciplinary, collaborative work on these basic questions.

Key Terms

Encoding-storage-retrieval (E-S-R) model: standard view of memory according to which information is first encoded, then stored, and then retrieved at recollection.

Engram or memory trace: vehicles that enable the preservation of memories from encoding to retrieval.

Standard Model of Memory (SMM): influential taxonomy of memory systems that divides long-term memory into declarative (explicit) and nondeclarative (implicit).

Episodic memory (EM): memory for specific spatiotemporal events; it is said to be accompanied by autonoetic consciousness (i.e., awareness of oneself as having experienced the event in the past).

Semantic memory (SM): memory for conceptual and factual information; it is said to be accompanied by noetic consciousness.

Consolidation: the stabilization of a memory trace that has been stored after encoding.

Content (in)variantism: the view according to which a memory's content at encoding is (different) identical to that memory's content at retrieval.

Representationalism: in the context of this chapter, the view according to which memory is mediated by representations.

1. INTRODUCTION

The word “memory” is polysemous. It can be used in reference to at least one of three things. First, it may refer to the psychological *faculty* by means of which we encode, store, and retrieve information. Second, it may refer to the *processes* of encoding, storing, and retrieving information. Or, third, it may refer to the encoded, stored, and retrieved *information* itself. Memory researchers have been interested in all three senses of the term, and their views on each often intersect.

1.1. Historical Overview

What is memory? Understood as a faculty, researchers—both philosophers and scientists—take memory to mean, at least *prima facie*, our capacity to bring back to mind information previously acquired in order to deploy it for present and future purposes. This definition is, of course, contentious; for some it may be too restrictive, while for others too permissive. Nevertheless, if there is something almost all memory researchers would agree on, it is that the term “memory” does *not* refer to a single monolithic psychological faculty but, instead, that there are different kinds of memory. What they disagree on, however, is what these kinds are and how to distinguish them.

The idea that there are different kinds of memory is old. Aristotle, for instance, famously distinguished between *memory* and *reminiscence*, the former roughly corresponding to the retention of temporally based information from past events, and the latter to the act of retrieving previously stored information (Chapell, 2017). This distinction was conceptual, and it was grounded in the theoretical apparatus of his theory of the mind. Yet, it had profound empirical consequences. He thought, for instance, that memory was perceptual, and thus certain animals had memory. By contrast, he took reminiscence to be a kind of inference, and, thus, only humans, with their intellectual souls, were capable of reminiscing. Augustine (*Confessions*, Book X) also talked about different kinds of memory. He argued, for instance, that there is a kind of memory that is *imagistic*, as when one remembers particular objects, but there is also a kind of memory that is *not*, as when one remembers mathematical laws. Locke (1694/1979) also distinguished two kinds of memory: *contemplation*, or the retention of consciously entertained information for short periods of time—roughly what we would call nowadays *working memory*—and *memory proper*, which he famously identified with the moniker “the storehouse of our

ideas." A similar distinction was later recapitulated by Russell (1921), who distinguished between *immediate* and *pure* memory, with only the latter pertaining the retention of ideas of temporally remote events no longer present in one's immediate memory. James (1890) offered another well-known distinction, between *primary* and *secondary* memory, with the former roughly corresponding to skills and habits and the latter to what the standard model calls *declarative memory* (see later discussion). This, of course, is only a small selection of the many kinds of memory that philosophers have offered in the past. Nevertheless, they all share something in common: they are established mostly on *conceptual* grounds. Their relationship to empirical evidence is, at best, tenuous.

In addition to wondering about the nature of memory, researchers have been interested in what we *do* with memory, namely, remembering. Across these various ways of making distinctions within memory, philosophers tend to agree with the idea that remembering is a *diachronic* capacity that requires (at least) two distinct events. There must be an initial event where one acquires information and also a subsequent event during which the previously acquired information is called to mind. According to Ryle, remembering is merely "having learned something and not forgotten it" (1949, p. 272). This is true enough, but many have wanted to say more. Even Ryle's minimalist construal involves reference to both learning and the later activity of not forgetting. Most memory theorists prefer to put the point more substantively: remembering is more than not forgetting, it is *retention*. The initial event or activity must be connected to the later event or activity through an extended process.

Cognitive psychologists and neuroscientists also agree that remembering is a diachronic process, but they tend to think that it involves three stages: encoding, storage, and retrieval. *Encoding* refers to the process by which information makes its way into memory. According to many theorists, one must actually attend to a piece of information in order for it to be a candidate for remembering. *Storage* refers to the process by which information is maintained in memory, and *retrieval* is the process by which information is recovered from memory. The **encoding-storage-retrieval**, or **E-S-R, model** of memory reflects the information-processing framework around which much of contemporary cognitive psychology is structured (Atkinson & Shiffrin, 1968). It is rare to find an introduction to memory that does not begin with a description of these three processes. Jenkins (1979) identifies these processes as essential features of memory experiments, and Schacter (2001) refers to them as the "pretheoretical orienting framework within which a systematic analysis of memory can be pursued" (p. 138).

Finally, both scientists and philosophers of memory have also wondered about the nature of memories per se. If remembering across its many forms is diachronic, then it is reasonable to assume that something must connect these processes across time. Researchers call this connecting factor by many names, including "memory representation," "**memory trace**," and "**engram**." The notion of memory trace has been central to discussions of memory for as long as this psychological capacity has been discussed. However, central aspects of the trace remain hotly contested, not only because the interpretation of memory traces varies from the metaphoric to the literal (Draaisma, 2000; Roediger, 1980) but also because, even when interpreted literally, their precise nature is unclear.

The idea that our experiences leave traces in our memory dates at least to Plato, who compared the process to that of seal rings leaving impressions on a wax table (*Theaetetus*, 1922). For Plato, these *impressions* needed to resemble the past events that they represented. Aristotle also thought that it was necessary for memory traces to be similar

to the events of which they were traces. However, he thought this condition was insufficient, as it may be possible to entertain a representation of a past event that is similar, or even identical, to such an event, without it having been acquired by the experience. As a result, he argued that for a mental representation to be a memory trace, it needed to meet three conditions: (1) to have been causally produced by the remembered event (causal condition), (2) to retain the content experienced from encoding to retrieval (retention condition), and (3) to be similar or isomorphic to the event experienced (similarity condition). Despite its ancient origin, Aristotle's notion of memory trace scaffolded many discussions in the philosophy of memory throughout the modern period and remained almost identical until the first half of the 20th century (Malcolm, 1977).

1.2. Historical Development

Despite shared origins, philosophical and empirical research on these questions—about the forms of memory, the process of remembering, and the nature of memories themselves—progressed independently. While philosophers and neuroscientists have generally agreed about the need to distinguish various forms of memory, to characterize each stage of the remembering process, and to account for the underlying memories, each field has focused on distinct methods, processes, and levels of explanation.

More recently, philosophers have gone beyond conceptual taxonomies of memory to also offer *grammatical* ones (e.g., Bernecker, 2010). According to this approach, distinct kinds of memory correspond to the different kinds of direct objects that the verb “to remember” can take on. Specifically, one can remember (1) particulars, such as objects or persons (e.g., “Cecilia remembers her childhood teacher”); (2) events and actions (e.g., “Edgar remembers falling down the stairs”); or (3) facts and propositions (e.g., “I remember that the Pythagorean theorem is $a^2 + b^2 = c^2$ ”). Unfortunately, to distinguish kinds of memories on grammatical grounds is problematic. For one, it is linguistically chauvinistic, as there are plenty of languages that do not conform to the syntactic structure of the English verb “to remember” (Amberer, 2007). Even if there were conformity, this grammatical structure would be unlikely to mirror psychological kinds, because it is perfectly possible to employ different grammatical forms to express the same thought. Harry may remember the event of meeting Sally and may express it as either “I remember meeting Sally” or “I remember that I met Sally,” without it meaning that in the former case he is expressing a different kind of mental content than in the latter. To take a grammatical difference as evidence for a difference in thought at best gets the order of explanation backwards.

Perhaps the most predominant taxonomy among scientists, and increasingly so among philosophers, is known as the **Standard Model of Memory (SMM)**. Initially suggested by Squire (1986) to account for neuropathological evidence, the SMM was solidified in the following years by incorporating further *empirical* findings from research on human and nonhuman animals (Box 21.1). Importantly, the SMM draws distinctions on the basis of *double dissociations*: roughly, the empirical demonstration that two psychological processes can occur independently of each other (see Chapter 2). The first such distinction is between short- and long-term memory, with the latter in turn splitting into declarative (or explicit) and nondeclarative (or implicit) memory (Figure 21.1).

Box 21.1 Standard Model of Memory and neuropsychological evidence

Due to intractable epilepsy, in 1953, at the age of 27, Henry Molaison or H.M. underwent a surgical procedure that removed both hippocampi and much of the surrounding tissue in the medial temporal lobes. As a result, H.M. lost his capacity to store and retrieve verbally reportable information whose acquisition required conscious awareness. By contrast, the surgery spared many of H.M.'s cognitive capacities, including perception, higher-order reasoning, and short-term memory, that is, the retention of information for brief periods of time (Scoville & Milner, 1957). Critically, it also seemed to spare the acquisition and retention of both perceptual and motor skills, which are acquired over time, need not be verbally reportable, and do not typically require conscious awareness to encode or retrieve (Milner et al., 1968). Thus, H.M.'s single dissociation case suggests that encoding in short-term memory can be preserved when encoding in long-term memory is impaired, and that nondeclarative or implicit memory can occur without declarative or explicit memory.

While H.M.'s case cannot provide evidence for double dissociation on its own, further studies provided the required evidence. For instance, Warrington and Shallice (1969) reported the case of K.F., who presented with severe short-term memory deficits due to a parieto-occipital lesion and yet had preserved long-term memory (i.e., did not exhibit anterograde or retrograde amnesia). Likewise, Gabrieli and colleagues (1995) reported the case of M.S. who, after a lesion in the right occipital lobe, showed impairments in implicit memory but not explicit memory. Additionally, Knowlton et al. (1996) showed that while individuals with medial temporal lobe amnesia (such as H.M.) were able to slowly learn probabilistic associations, individuals with damage in the basal ganglia and striatum—such as patients with Parkinson's or Huntington's disease—could not. By contrast, basal ganglia damage did not impair performance on recognition tasks, precisely the tasks on which individuals with amnesia were profoundly impaired.

Finally, research with neurotypical populations, as well as comparative studies with nonhuman animals, helped to solidify the idea that long-term memory comprised two main kinds of memory: declarative or explicit and nondeclarative or implicit (Squire, 1992). Further refinements divided the latter into four subclasses, each associated with a distinct neuroanatomical substrate. Priming—or the increased propensity to generate a response given the implicit processing of a previously presented stimulus—was associated with the neo-cortex; procedural memory (i.e., memory for skills and habits) became associated with the basal ganglia/striatum, the motor cortex, and the cerebellum; emotional associative learning was associated with the amygdala; and nonassociative learning, such as reflexes, was associated with neural reflex pathways.

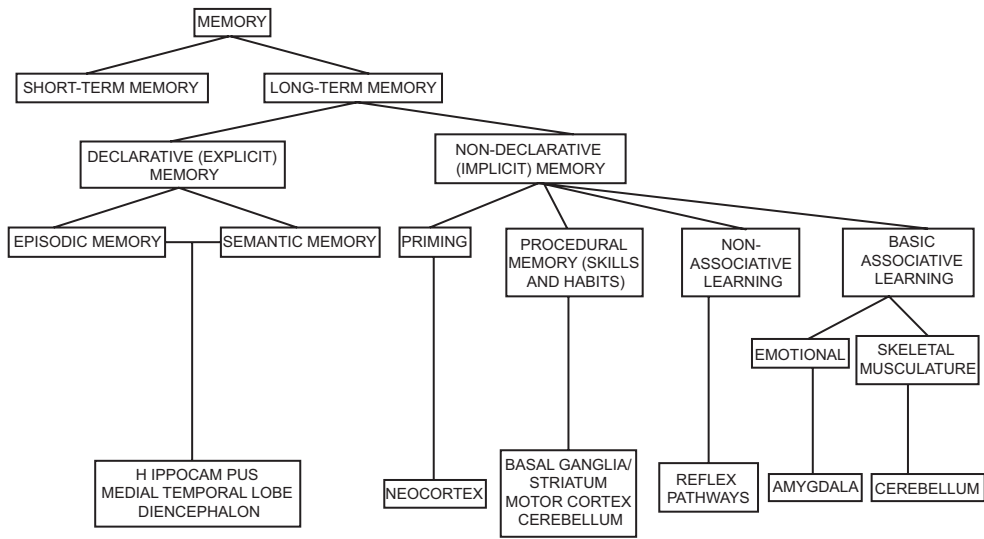


FIGURE 21.1 Memory

Declarative or explicit memory is often divided into two subclasses as well, thanks to Tulving's influential distinction between **episodic memory** (EM) and **semantic memory** (SM) (Tulving, 1972). Initially, this distinction was proposed on the basis of putative computational differences between the two. The first difference had to do with the nature of the stored information: EM stored information according to spatiotemporal relations, while SM stored it according to semantic associations. Second, the information in EM was autobiographical and was recorded directly (i.e., the information recorded had to be experienced), whereas in SM the information encoded need not be autobiographical and it could be recorded indirectly. Third, EM was prone to interference and modification in a way that SM was not. A few years later, Tulving (1985) refined this definition and introduced a phenomenological criterion to distinguish the two: EM involved *autonoetic* consciousness (self-knowing), while SM involved *noetic* consciousness (knowing)—procedural memory was thought to involve *anoetic* (nonknowing) consciousness. What exactly these terms mean, and how these kinds of consciousness relate to the different kinds of memory, is still a matter of debate. Nevertheless, Tulving's influential proposal is that EM and SM constitute two distinct kinds of memory, with independent computational and phenomenological characteristics.

With regard to the process of remembering, philosophers and psychologists have shared a commitment to the E-S-R approach, but have focused their work on different ends of the process. For memory scientists, most of the interest in memory processes has been concerned with encoding. In a 2000 paper, Roediger estimated that 80% of experiments in the cognitive psychology of memory had been focused on encoding. Researchers vary the conditions of encoding and observe their influence on subsequent retention. In neuroscience, much of the focus is on the process of **consolidation**: the encoding process by which information moves from temporary to long-term memory

storage. The observation that newly acquired information takes a while to solidify into a long-term memory has been noted by many, including the Roman rhetorician Quintilian (IC AD/1921) and Ribot (1882), a doctor working with amnesic patients over a century ago. In contemporary memory science, most distinguish between two forms of consolidation, each with a distinct timescale (McGaugh, 2000). First, there is *synaptic consolidation*, which is the initial stabilization of the memory trace or engram that occurs directly after learning (i.e., within the first few minutes or hours). The process of synaptic consolidation, in turn, is understood as the process by which these strengthened connections are stabilized so as to make the engram long-lasting. Without synaptic consolidation, the strengthened connection between neurons that occurred during encoding will weaken and the memory will be lost (Weiler et al., 1995). Second, there is *systems consolidation*, which occurs at some time point after synaptic consolidation, where the memory moves from the hippocampus to the neocortex. The shifting of memories from one system to another is thought to aid in overall memory storage by preventing the hippocampus from becoming too overloaded with memories. The precise timescale for systems consolidation is a matter of some debate; it is thought to occur anywhere from a few weeks or months to possibly even years or decades after the initial encoding.

Philosophical accounts of remembering have often focused on the other end of the remembering process, namely, retrieval and the concomitant experience of remembering. Specifically, philosophers have emphasized the difference between the content brought to mind in remembering and the content entertained during the initial experience. However, this approach introduces the challenge of characterizing the precise relationship between the contents of the encoded and the retrieved representations. **Content variantism** holds that the content of the retrieved representation differs from that of the encoded one. Throughout history, there have been a range of views on how precisely they differ. Some philosophers suggested that the retrieved content is accompanied by an additional *cognitive* state, such as the belief that the remembered event happened in the past (James, 1890; Russell, 1921). Others held instead that the additional state was *affective*, such as an emotion or feeling of pastness (Spinoza, 1677/1985). Finally, for other philosophers, the retrieved content is accompanied by a second-order representation or *apperception* of the encoded content (Leibniz, 1714/1991).

In contrast, preservationist views of remembering often advocate for content invariance, according to which the content of the encoded representation is identical to that of the retrieved one (Zemach, 1968). The most prominent form of preservationism has been the Causal Theory of Memory. The basic commitment to remembering as a causal process is widespread; support can be found from Aristotle through to Ayer (1956). The official Causal Theory of Memory, however, is generally attributed to Martin and Deutscher (1966). The view, which has been refined and extended by others more recently (e.g., Bernecker, 2010; Debus, 2010; Perrin, 2018), involves commitment to remembering as a causal process, as well as an additional set of criteria that helps to identify which causal relations between the past event and its subsequent activation count as remembering.

The divide between philosophical and empirical approaches to memory continues in research on the memory trace. There is a shared commitment to the existence of some

such trace, but philosophers have focused primarily on its content, while neuroscientists have focused on the underlying mechanism (De Brigard, 2020). On the philosophy side, the debate has focused on how to interpret three putative conditions for an entity to count as a memory trace, which derive from Aristotle's account as described previously: (1) the *causal condition*, whereby a memory trace must play a causal role in the recollection of the memory of which it is a trace; (2) the *retention condition*, according to which a memory trace must retain the experienced content from encoding to retrieval; and (3) the *isomorphism condition*, according to which a memory trace must structurally resemble that which it represents (De Brigard, 2014a).

Philosophical theories about memory trace content can be distinguished by how they characterize the relationship between the content in the initial perceptual experience and the content in the remembering of that experience. There are three general approaches. According to *direct representationalism*, memory traces are representations created only after perception. In this view, perception is unmediated and representations of the object to be remembered are formed after this initial sensory act. *Semidirect* representationalism, in contrast, characterizes perception as mediated by a representation. This representation persists and is reactivated in remembering. In other words, the memory trace is the very same representation created during the initial perception and then retrieved during recollection. Finally, *indirect* representationalism takes perception to be mediated by a representation, as semidirect representationalism does, but it takes this representation to be numerically distinct from the representation that serves as the memory trace. The representations may carry the same content but are not the same tokens of that content.

On the science side, memory researchers have long been dedicated to the search for the cognitive and neural mechanisms that make remembering possible (Box 21.2). In a pioneering work, the German biologist Semon (1921) coined the term **engram** in the early 20th century, refashioning the age-old memory trace as the neurological process by which information is encoded, stored, and retrieved. Semon's work was cut short by his early death. Work on engrams languished afterward and was challenged by trenchant skepticism from Lashley (1933) and others. The engram's revival began as Hebb's work on neural organization became applicable to memory. Hebb's suggestion was that the strength of the connection between neurons—the synapse—changes as a function of the interaction, making it easier for these neurons to reactivate together, thereby facilitating the retrieval of information encoded during the initial experience (Hebb, 1949). This, in turn, led to the discovery of long-term potentiation as the more precise mechanism for strengthening synaptic connections in memory (Bliss & Lømo, 1973; Craver, 2003). Hebb's theory of neural communication, coupled with the discovery from Milner and colleagues that the hippocampus was the neural structure supporting long-term memory (as discussed in Section 2), led neurobiologists to look for these structural changes in the synaptic connectivity between hippocampal neurons. Our understanding of how hippocampal neurons communicate and change so as to support retention has been refined and revised over time, but nonetheless it remains the case that "the prevailing view is that the formation of an engram involves strengthening of synaptic connections between populations of neurons that are active during encoding" (Josselyn et al., 2015, p. 421).

Box 21.2 Neural mechanisms of episodic memory

The case of H.M. evidenced the critical role the hippocampus plays in episodic memory. Located in the middle of the temporal lobes, the hippocampus is a structure of ~3.5 cm³ of archicortex (archi = old). Unlike the neocortex, which comprises six cortical layers, the hippocampus presents only three, and neuroanatomically it is divided into the dentate gyrus, the cornu ammonis (CA) fields (CA1, CA2, and CA3), and the subiculum. Along with the adjacent pre- and parasubiculum and the entorhinal, perirhinal, and parahippocampal cortices, these structures form the medial temporal lobe (MTL).

According to the SMM, when an event is encoded, it undergoes an initial synaptic consolidation in the hippocampus, whereby the information is stabilized in a process that lasts from a few minutes to 24 hours (straight arrow in Figure 21.2). This rapid, synaptic consolidation can be interrupted, and its disruption via drugs or trauma can lead to memory loss. Neurophysiologically, synaptic consolidation critically depends on long-term potentiation, which contributes to the strengthening of synapses and to neural plasticity (see Chapter 3). Information that has been consolidated at the synaptic level in the hippocampus then undergoes a second, systems consolidation, whereby the different components of the episode are stored in the regions of the neocortex that initially process them (curved arrows in Figure 21.2). Finally, at retrieval, the prefrontal cortex is thought to orchestrate the reactivation of the relevant portions of the neocortex, leading to the reinstatement of the encoded memory (dotted arrows in Figure 21.2). This process is thought to be independent of the hippocampus, which would explain why damage in the hippocampus leads to anterograde amnesia—that is, the incapacity to encode new episodic memories—but not to total retrograde amnesia, that is, the incapacity to retrieve episodic information acquired prior to the cerebral insult.

Recently, however, the standard view of consolidation has been challenged on a number of fronts. One such challenge comes from the observation that the amount of remnant hippocampal tissue seems to negatively correlate with the extent of the retrograde amnesia, so that individuals with damage to the entire hippocampal complex suffered complete amnesia regardless of the age of the past memories. This suggests that either those memories were never fully consolidated, despite their age, or that the hippocampus is still required for memory retrieval, which both go against the traditional view of consolidation from the SMM. The *multiple trace theory* (MTT) has been proposed to account for this and other conflicting counter-evidence, and it holds that although the prefrontal cortex may initiate the retrieval effort, the hippocampus is still required to help to piece together the information distributed in the neocortex (Nadel & Moscovitch, 1997; Moscovitch et al., 2005). Another challenge to the standard view from the SMM comes from research on reconsolidation, according to which, upon retrieval, memories become labile and prone to modification. These findings, once again, suggest that processes akin to those of synaptic consolidation occur at retrieval, further suggesting that the MTL is engaged both at encoding and retrieval of episodic information.

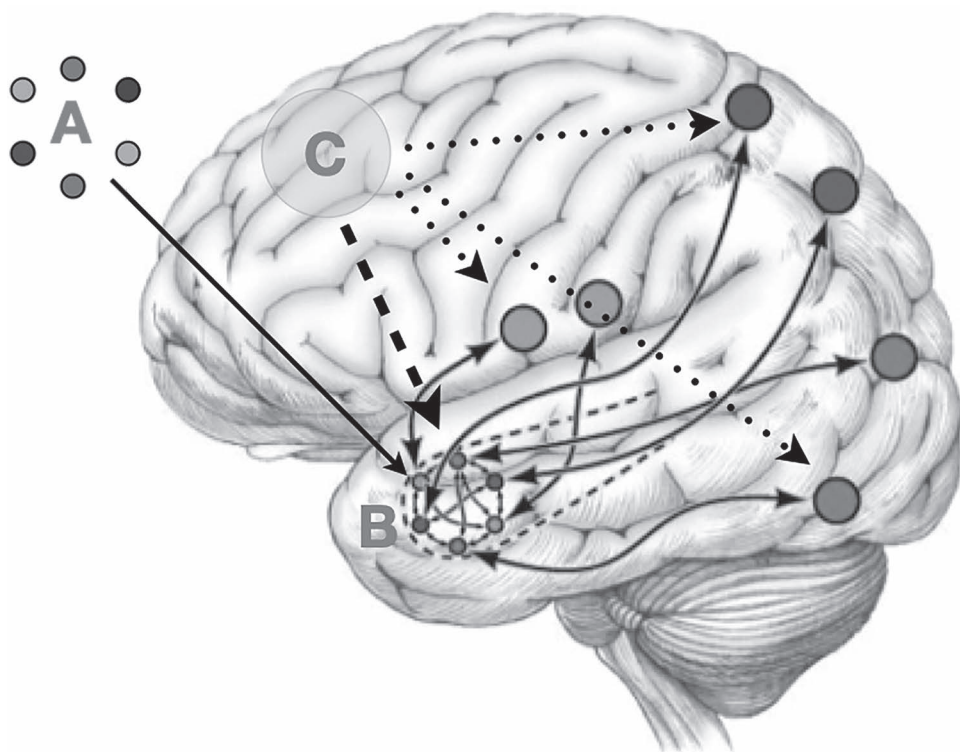


FIGURE 21.2 Memory consolidation and retrieval: (A) encoded event (shaded circles represent sensory information); (B) approximate location of the hippocampus and surrounding medial temporal lobe regions: (C) prefrontal cortex, thought to initiate retrieval upon cue.

Work on the memory trace in cognitive psychology has largely complemented the search for its neural mechanism. In fact, it provides the functional characterization that guides the search for the requisite neural structures. If remembering is a process of encoding, storage, and retrieval, then the memory trace is the item that unites these three stages—that is, the trace is what is encoded, stored, and then retrieved. Indeed, this is exactly how Tulving more recently characterized remembering:

An event happens, a person experiences it, memory traces are laid down representing the event, the past vanishes and is replaced by the present. The memory traces of the event continue to exist in the present, they are retrieved, and the person remembers the event. This, in a nutshell, has been the understanding of how memory works. It is simple and straightforward; there is no need or room for magic, or marvel.

(Tulving, 2002, p. 19)

2. CONTEMPORARY ISSUES

The story of the philosophy and neuroscience of memory, as told earlier, is of two independent research trajectories. In recent years, however, the concerns of these fields have begun to intersect, resulting in more collaborative and interdisciplinary work.

On the question of memory systems, philosophers have noticed that several empirical dissociations captured by the SMM seem to mirror conceptual distinctions well-known to them. For instance, some argue that the distinction between declarative-explicit and nondeclarative-implicit memory reflects the traditional distinction between know-that and know-how (Ryle, 1949). As a result, some philosophers have used the SMM to argue not only for a conceptual but for a psychological distinction between the two kinds of knowledge (Wallis, 2008; Devitt, 2011; Glick, 2011; Brown, 2013). By contrast, others have used concerns about the SMM to argue against the claim that know-that and know-how are psychologically different kinds of knowledge (Stanley & Krakauer, 2013; De Brigard, 2019).

The distinction between EM and SM also seems to coincide with conceptual distinctions drawn by philosophers (e.g., Augustine). However, there is still much disagreement as to what makes EM distinct from SM. One possibility is to characterize them functionally, in the spirit of Tulving's initial proposal. Accordingly, it has been suggested that EM processes spatiotemporal information about *what*, *when*, and *where* something occurred. Such characterization separates EM from reportability and awareness and thus, in principle, enables the possibility of identifying EM in nonhuman animals (Clayton & Dickinson, 1998). But others disagree, arguing not only that SM could suffice to provide *what-when-where* information but also that such characterization misses an essential feature of EM: its peculiar phenomenology, that is, the fact that it feels like mentally traveling to one's past. Unfortunately, understanding the psychological requirements of this experience has proved elusive. Some suggest, for instance, that EM requires the capacity to deploy temporal concepts (McCormack & Hoerl, 2008), or a sense of self (Rowlands, 2016), or even a noninferential connection with autonoetic consciousness (Klein, 2015).

A related issue is that of distinguishing EM from imagination. Historically, both philosophers and scientists have acknowledged critical similarities between the two faculties. For instance, both involve representing objects or events that need not be present. But what exactly distinguishes them is still a matter of debate. For some, the difference between memory and imagination lies in something *intrinsic* to their content, such as vivacity or associated cognitive or affective states. Unfortunately, these views are fraught with problems, such as the fact that some imaginings can be experienced more vividly than certain memories, or the fact that one may associate memory beliefs or feelings to contents one only imagined. Alternatively, some argue that the difference between memory and imagination is *extrinsic* to their contents, as memories—unlike imaginations—need to be true and/or should be causally related to the event they are about. Yet, once again, these options face difficulties, not only because the precise causal relationship between the encoded and remembered event is difficult to characterize but also because memories are often false (for a review, see De Brigard, 2017). As a result, many contemporary views reject the claim that memory and imagination are two different faculties and suggest instead that memory is a subclass of a more general cognitive capacity to generate mental simulations (Schacter & Addis, 2007; De Brigard, 2014b; Michaelian, 2016).

Just as the distinctions between forms of memory have begun to break down, recent work on the process of remembering has brought challenges to our understanding of both ends of the process—encoding *and* retrieval. On the encoding side, the idea that synaptic consolidation brings permanence has been increasingly undermined by a steady line of research, which indicates that memories undergo a new period of vulnerability every time they are activated (Sara, 2000). Even though this mechanistic process is not identical to the initial consolidation, it has become known as *reconsolidation* (Nader et al., 2000). Now, instead of viewing consolidation as a one-time, finalizing activity, neuroscientists think that “consolidations never end” (Dudai, 2012, p. 227). The impact of reconsolidation on our understanding of remembering goes beyond changes to our characterization of its cellular and molecular mechanisms: it contributes to the increasingly popular view of remembering as a dynamic process (Silva, 2007). Systems consolidation, too, is no longer the received view of long-term memory preservation. In the previous section, we noted the persistence of debates over its time course. This has been taken by many as reason to question whether systems consolidation in fact occurs (Box 21.2).

At the other end of the process, philosophical accounts of remembering have also faced numerous challenges. Content variantists have offered various proposals for the “something else” that accompanies the previously stored information in the act of remembering. It is now well established that none of these proposed *memory markers* can adequately capture all, and only, the relevant cases (Bernecker, 2010). Worse, these various accompaniments to retrieved information, however characterized, appear to be woefully unreliable. Decades of empirical work into the pervasiveness of memory errors reveal that the feeling of remembering is fully separable from the successful retention of information. Across a range of conditions, it is continually demonstrated that people can be led to think that they are remembering when they are not (Loftus, 1997; Roediger & McDermott, 1995). In response, many philosophical accounts of remembering now rule out the need for any particular feeling, judgment, or explicit endorsement of what is retained (Martin & Deutscher, 1966), although there are some who continue to advocate for this as an essential component (e.g., Debus, 2018). Either way, content variantism involves a violation of the idea that remembering is focused exclusively on retention, and understanding the precise degree of variation is the topic of current investigations (Michaelian, 2016; Bernecker & Michaelian, 2017).

These challenges have contributed to a widespread shift in how remembering is characterized, from a process of *retention* to a process of *reconstruction*. It is not entirely clear what is meant by reconstruction, nor whether everyone who endorses this shift in terminology has the same conception in mind. But generally, those who favor reconstruction deny or downplay the role of explicitly stored information in memory and highlight instead the information and activities that occur after the event being remembered and at the time of recall.

Changes to the process of remembering bring with them changes to the idea of what is stored in memory across this process. Many philosophers of memory continue to discuss and appeal to memory traces, although it is now common practice to challenge the aforementioned conditions. The most common form of resistance is to the conception of memory traces as localized in a particular representation or mechanism in the brain. Sutton (1998) pioneered this line of work on memory traces, arguing for a view of traces as distributed and dynamic in philosophical work on memory dating back to the 17th

century. There are now many proposals for distributed traces on offer (e.g., Michaelian, 2011; De Brigard, 2014a). Other philosophers have argued that the contents of memory traces can be extended beyond one's brain (Clark & Chalmers, 1998) and even be distributed across people (Sutton, 2010). There are also theorists who have suggested that what makes a memory trace is not a categorical but a dispositional property (Robins, 2012; De Brigard, 2014a, 2017). And, consistently, many currently argue that in order to fully understand the nature of memory traces and the veracity of their conditions, we need to turn to research in the cognitive psychology and neuroscience of memory.

While conceptions of the memory trace are becoming increasingly less traditional in philosophy, advances in cellular and molecular neuroscience have led many in this area to believe that the engram, in the classical sense, has finally been found (Josselyn et al., 2015). The transition from early work on the neural mechanisms of memory to the current place of the engram in memory science is the result of a steady trajectory of experimental discovery across many areas and levels of neuroscience (Silva et al., 2014). These advances have been made possible by the development of new experimental tools: the use of transgenic model organisms, photon microscopy and other imaging techniques, and, most recently and most significantly, optogenetics. Optogenetics is an intervention technique that allows for the control of neural activity with light (Deisseroth, 2011). It has been used to provide the first model of false memory in nonhuman animals, to change the valence of memories (from positive to negative and vice versa), and to recover memories that were previously assumed to have been lost (Lui et al., 2012; Redondo et al., 2014; Ryan et al., 2015). These results expand, enhance, and in some respects challenge assumptions about the nature of memory and its underlying mechanisms. For some, these advances in the neurobiology of memory provide support for a fully reductionist view of memory, where the capacity is understood entirely in terms of the operations of these neural processes (Bickle, 2006, 2020; see also Chapter 3). Those who support this reductionist picture often speak in terms of engrams exclusively, setting aside the term memory trace and associated concerns about content, representation, and subjective experience.

3. FUTURE DIRECTIONS

As the science and philosophy of memory continue to develop and interanimate, it is difficult to anticipate trends for future research. Current discussions, however, offer glimpses of what may come. Our chapter began with the questions that shaped early inquiry—*What is memory? What is remembering? What are memories?* As research continues and interdisciplinary connections deepen, these questions are all being asked anew.

First, on the question of memory types, it seems likely that the SMM will be rejected. More and more empirical evidence is strongly suggesting that these allegedly distinct memories are profoundly intertwined, both psychologically and neuroanatomically, and that the traditional distinctions may not be carving nature at its joints. Indeed, recent proposals have offered alternative taxonomies based either on the nature of memory representations (Reder et al., 2009) or on their processes (Henke, 2010; Cabeza & Moscovitch, 2013). A related line of research pertains to the relationship between memory and other cognitive processes, including not only imagination but also perception and working memory. Despite the apparently obvious fact that they are distinct, there is much evidence

of neural overlap and reuse across these structures, suggesting that our reconceptualization of memory may have to go beyond memory itself and into other cognitive capacities (Barense et al., 2010).

These changing ideas about memory's systems and relation to other capacities impact, too, our conception of remembering. Researchers are rethinking our understanding of memory's function. The traditional view held that the function of memory was to remember the past, understood as encoding and storing a faithful representation of a past event in order to retrieve it with fidelity (Boyle, 2019). With retention no longer central to the process of remembering, this view is now contested. If retrieval is reconstructive, then how should we understand the function of remembering? A range of alternative functional hypotheses have been suggested, including the idea that memory is primarily forward looking, in that its primary role is to enable simulations of possible future events (Suddendorf & Corballis, 2007), or that its primary role is to strengthen social relations (Alea & Bluck, 2003) or even to curb future decision-making (Boyer, 2008).

All of these shifts make the long-standing commitment to the memory trace more uncertain than ever. Moving forward, our progress in understanding memory traces will follow the distinct lines of inquiry brought on by the development and use of new investigative techniques in neuroscience and the theoretical exploration of the reasons and requirements for traces among philosophers of memory. In addition, any significant advances will require a rapprochement between these divergent approaches to the underlying mechanisms.

Neuroscientists will need to explore how far optogenetics can go in allowing us to identify and manipulate memory traces. The technique thus far has been exciting because it offers a way to intervene into the memory trace directly, circumventing the retrieval process. In this way, it provides a novel workaround for many methodological challenges of memory research (Robins, 2016, forthcoming). How this will change experimental practice and how it will be further enhanced by the development of other techniques have become topics of discussion, among both philosophers of science (Sullivan, 2016) and neuroscientists (Josselyn et al., 2015). On the other end, as philosophers have jettisoned more and more of the traditional conditions on memory traces, there is renewed exploration of why there has been so much agreement on the need for a trace to begin with. Can we have an account of memory without it? Several recent trace-free proposals are on offer (Hutto & Peeters, 2018; Michaelian, 2016). Others find it difficult to jettison the concept entirely (De Brigard, 2020) and make the kinds of modifications discussed previously. Unfortunately, there is little agreement about which conditions to retain and which to let go.

Summary of Key Ideas

These are revolutionary times for the science of memory. Extant findings are debunking traditional models, and recent technological advances are enabling us to study memory processes at increasingly diverse temporal and spatial resolutions. Thankfully, philosophy of memory has taken notice and is undergoing a complementary revolution of its own, revisiting traditional distinctions, reevaluating long-established assumptions, and rebuking tacitly accepted conditions. The time is

ripe, then, for closer interaction between the science and the philosophy of memory. In this chapter, we explored a few avenues for pursuing this development. Specifically, we explored questions about what memory is and how many kinds of memory there are, as well as the process of remembering and the precise nature of the information stored in memory. Given memory's central role in cognition and its place in our social, emotional, and moral lives, questions about memory are likely to persist and expand as the joint exploration of this capacity continues in philosophy and neuroscience.

Discussion Topics

- Does remembering require memory traces?
- What is the difference between memory and imagination?
- Do memory traces need to respect all three putative conditions, namely, the causal, the retention, and the isomorphism conditions?
- How should developments in neuroscience affect the philosophical view of memory traces?
- Is the Standard Model of Memory an accurate representation of our memory systems? Can you think of types of remembering that are not captured by this model?

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